Geophysical Studies of North African Cenozoic Volcanic Areas I: Haruj Assuad, Libya

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This study is the first of a series in which a main concern will be to establish the Neogene absolute motion of the African Plate. A combined palaeomagnetic and K–Ar whole rock age dating study has been made of the Haruj Assuad basaltic volcanic area of central Libya. Seventy volcanic units, largely pahoehoe basaltic flows, were sampled in the north-central Haruj Assuad in an area centred on 27°45′ N, 017°30′E. Precisely defined paleomagnetic directions were obtained for 68 of the volcanic units and a weighted mean pole at 83°N, 171°E ($\delta p = 5°$, $\delta m = 9°$) obtained. This pole is just significantly different at the 95% level from the geographic pole. The difference between the paleomagnetic and geographic poles is thought to be largely the result of a degree of nonrepresentative sampling of the geomagnetic field. K–Ar radiometric age data indicate that volcanic activity in the Haruj Assuad area continued at least over the interval 6.0 to 0.4 my. The polarities of dated units are in agreement with the predictions of the time polarity scale. The ages of dated flows indicate that most of the flows presently exposed are younger than 2.2 my (*i.e.*) Upper Pliocene and younger. A scheme for assigning relative ages to groups of flows by degree of surface weathering is now calibrated in part with absolute ages.

Cette étude est la première d'une série qui a pour but principal de déterminer la nature du mouvement absolu Néogénien de la Plaque Africaine. La détermination du paléomagnétisme et la datation par la méthode K–Ar de la roche totale ont été faites sur les volcaniques basaltiques de Haruj Assuad de la région centrale de la Libye. Soixante-dix unités volcaniques, principalement des coulées volcaniques pahoehoe, furent échantillonnées au centre nord de Haruj Assuad dans une région centrée sur les coordonnées 27°45′N et 017°30′E. Des directions très précises de paléomagnétisme ont été obtenues pour 68 des unités volcaniques ainsi qu'un pôle, moyen à 83°N, 171°E ($\delta p = 5°$, $\delta m = 9°$). Ce pôle est significativement différent du pôle géographique au niveau de 95%. Cette différence est le résultat du degré d'échantillonnage non représentatif du champ magnétique. Les données des âges radiométriques K–Ar indiquent que l'activité volcanique dans la région de Haruj Assuad a été continue au moins pour la période de 6.0 à 0.4 ma. Les polarités des unités datées sont en concordance avec les prédictions de l'échelle des polarités. L'âge des coulées datées indique que les coulées présentement exposées sont plus jeunes que 2.2 ma, *i.e.* Pliocène supérieur et plus jeune. Un modèle qui consiste à donner les âges relatifs de groupes de coulées en fonction du degré d'altération de surface est maintenant calibré en partie avec les âges absolus. [Traduit par le journal]

Introduction

Recently considerable interest has been shown in the absolute motion of the African Plate. Burke and Wilson (1972) suggest that the African Plate has been essentially stationary for the last 25 my, while Morgan (1972) has determined a typical major-plate magnitude of motion for the Plate. The absolute motions of plates can be obtained only from plume expressions, or in the latitudinal sense, from palaeomagnetic data. This series of papers will give palaeomagnetic and radiometric age data

for the line of volcanic areas extending for 1400 km from the Mediterranean littoral close, to Tripoli in Libya, to Tibesti in Chad (Fig. 1, inset). The linear nature of this chain of volcanic areas suggests a plume trace origin. A test of the reality of this type origin for the volcanic line should give evidence on the Cenozoic motion of the African Plate.

This first paper of the series describes data from the Haruj Assuad basaltic volcanics of central Libya. During this century, Kanter (1940), Bellair *et al.* (1952), Klitzsch (in Williams 1966) and Klitzsch (1967) have given geological descriptions of this relatively inaccessible area. Klitzsch (1967) gives the most comprehensive account of the area, based on information gathered during oil exploration activities. The 45 000 km² volcanic area is the largest in North Africa, but is generally of low elevation and relief, suggesting formation by a relatively small number of very fluid pahoehoe basaltic flows. Occasional cones, shield volcanoes and plugs interrupt the monotonous scenery. The frequent preservation of volcanic landforms suggests a recent termination for volcanic activity in the area. Stratigraphic evidence indicates that the age span of volcanic activity is Middle Miocene and younger (*i.e.* <18 my). Klitzsch (1967) has recognized six major groups of flows of successively younger Pliocene to Holocene age (B1 to B6/7) from the degree of weathering of flow surfaces. It is suggested by Klitzsch that volcanic activity in this area was related to three major tectonic elements of very different age: the early Palaeozoic southern Haruj uplift, the western edge of the Mesozoic Tibesti–Sirte uplift, and the southeastern prolongation of the Hon graben fault system of late Cretaceous to Tertiary age. Overlap of eruptive activity centered on these three features is supposed to account for the great area covered by volcanics.

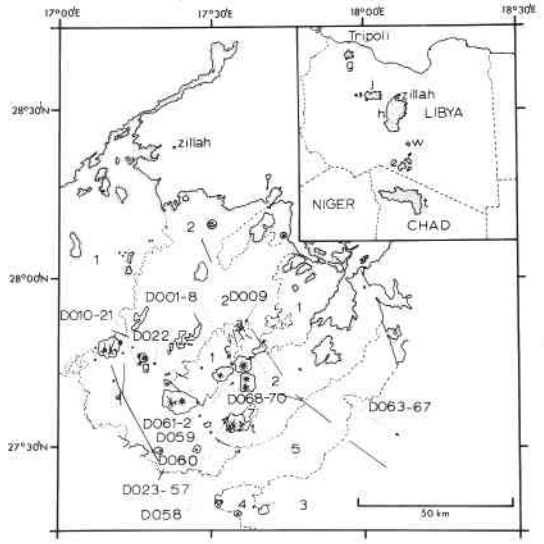


FIG. 1. Location of Haruj Assuad sampling sites. Main map: Numbers 1, 2, 3, 4, 5, and 7 refer to age groups of flows as identified by Klitzsch (1967). D001–8, etc.: Volcanic units sampled at different sites (see also Table 1)
(a) airstrip for 1970 Prakla camp
(c) 1970 Prakla camp
(g) Garet el Gazia cone
(t) Gleb et Tleah shield volcano
Note: Zillah is also spelt Zella
Inset: Cenozoic volcanic areas of north central Africa
(g) Garian
(j) Jabel Soda
(h) Haruj Assuad
(w) Wau en Namus
(e) Eghei
(t) Tibesti

Sampling

Sampling was carried out using portable diamond drills of the USGS type. Seventy volcanic units were sampled, at four drill cores per unit, within a radius of 30 km of the Prakla G.m.b.H. seismic exploration camp then situated in the north-central part of the Haruj at close to 27°45′N, 17°30′E (Fig. 1). The units sampled were mainly scoriaceous basalts of the pahoehoe type and were chosen to give as wide range of coverage as possible of the age groups of lavas described in Klitzsch (1967) (Table 1). Samples were obtained from Groups B1, B2, and B5 and from a cone, a plug, and a caldera within a shield volcano. These latter volcanic features have not been classified by degree of weathering.

Natural Remanent Magnetism of the Haruj Assuad Volcanics

The natural remanent magnetism (NRM) of 2.5 cm long, 2.5 cm diameter measurement samples cut from drill cores was measured using a parastatic magnetometer. All samples were subjected to systematic alternating field remanence cleaning in steps of 50 Oe until final directions were evident. Unit mean cleaned

TABLE 1. Location of Sections (see also Fig. 1)

Section	Units	Location
1	D001–5	Stratified flows in walls of large wadi (D005 youngest). (B1 age group of Klitzsch 1967).
2	D006/7	Young flows resting on the top of D005. Original surfaces of flows preserved.
3	D008	Blockwerk or Aa type of flow. Pahoehoe flow D007 appears to have flowed around the nose of D008.
4	D009	Scoriaceous lava forming several small rounded hills. (Possibly age group B2 of Klitzsch 1967).
5	D010–21	In walls of Garet el Gazia cone (D021 youngest).
6	D022	Vesicular basalt flow about 3 km east of Garet el Gazia (age group B2 of Klitzsch 1967).
7	D023–57	Sequence of thin flows in wall of westernmost Gleb et Tleah caldera (D057 youngest).
8	D058	At edge of prominent flow (age group B5 of Klitzsch 1967).
9	D059	Prominent blockwerk or Aa plug.
10	D060	Scoriaceous flow at foot of plug D059.
11	D061/2	In walls of wadi (D061 younger than D062). (Age group B2 of Klitzsch 1967).
12	D063–7	Flows with original topography largely preserved.
13	D068–70	Blockwerk or Aa flows(?) in a gully in the prominent north facing cliff of the Garet Umm el Asol massif.

remanence directions were obtained by a computer search for the minimum scatter between the directions for different samples (Dagley and Ade-Hall 1970) preceded where necessary by the removal of clearly erroneous data. The unit mean cleaned directions are listed in Table 2. We note that it was impossible to obtain a satisfactory mean direction for only two units. The first of these, flow D013, belongs to the section forming the outer part of the southern wall of the Garet el Gazia Cone. Several flows in this sequence had anomalously strong NRM intensities strongly suggesting isothermal (or thermal) remagnetization resulting from lightning strike. With the exception of flow D013, it was possible to reject this anomalous data while still leaving sufficient information to obtain a physically meaningful mean direction. For D013 it is only possible to quote a reverse NRM polarity with confidence. The other unit failing to yield a satisfactory mean direction was plug D059. This plug forms a prominent geographic feature like the Garet el Gazia cone and the anomalously strong NRM's again suggest lightning induced magnetization. However, the NRM of this intrusive rock is much softer than for most of the flows and no consistent original magnetization direction could be obtained. The precision of definition of the mean directions of units was high, with 90% of the units having α_{95} angles of less than 5° associated with their mean cleaned NRM directions.

The first feature of interest in the palaeomagnetic data is the widespread occurrence of both polarities. Our rather limited data show that age group 1 of Klitzsch (1967) contains R polarity flows, age group 2 both R and N flows, and age group 5 (one flow), N polarity. Within section NRM directional variation is sufficiently restricted that the groups of directions for different sections are quite distinct (Fig. 2). The data from section 7, which consists of 35 thin flows forming a caldera wall, is of particular interest in this respect, both in the evaluation of the secular variation at the time of eruption of the sequence, and in assessing the precision of palaeomagnetic data required for a secular variation study. Figure 3 shows the data for this section plotted against unit number, giving an imposed appearance of temporal regularity of eruption. An apparently systematic variation of declination and inclination is observed throughout this eruptive episode, with respective ranges of 25° and 10°. The least variation from flow to flow occurs for the central part of the sequence (D029 to D051). Here and elsewhere we may be looking at genuine periods of low secular variation but a viable alternative is that the rate of extrusion of flows was highest during the central interval of the lifespan of the exposed part of the caldera. This latter interpretation is supported by the absence of weathering of flow tops and of interflow sediments. The individual flow directions fit satisfactorily to a smooth curve where α_{95} is less than 3°. Where α_{95} is greater than 3°

TABLE 2. Mean directions of cleaned natural remanence for Haruj Assuad volcanics

Unit number	D(°)	I(°)	k†	α_{95}(°)†	Number used (Number sampled)
Section 1					
A10D001	181	−29	1300	1.5	4(4)
2	182	−29	516	2.4	4(4)
3	207	−35	2239	1.2	4(4)
4	205	−35	400	2.8	4(4)
5	199	−34	247	3.5	4(4)
Section 2					
6	011	+38	197	4.8	3(4)
7	003	+27	809	1.9	4(4)
Section 3					
8	181	−21	765	2.4	3(4)
Section 4					
9	177	−46	23	11.9	4(4)
Section 5					
10	183	−38	2720	1.3	2(4)
11	171	−35	389	3.4	3(4)
12	179	−30	589	2.3	4(4)
13	No satisfactory mean direction (R)				
14	180	−39	858	2.4	2(4)
15	189	−39	81	6.2	4(4)
16	170	−37	318	3.1	4(4)
17	170	−40	112	5.2	4(4)
18	190	−40	211	4.6	3(4)
19	172	−36	503	2.5	4(4)
20	172	−36	3000	1.3	2(4)
21	173	−33	400	2.9	3(4)
Section 6					
22	340	+70	1200	1.3	2(4)
Section 7					
23	184	−47	996	1.5	4(4)
24	178	−57	426	2.7	4(4)
25	182	−54	392	2.8	4(4)
26	181	−61	129	4.9	4(4)
27	187	−52	931	1.8	4(4)
28	192	−48	218	6.0	3(4)
29	196	−54	400	2.9	3(4)
30	194	−52	694	2.1	4(4)
31	193	−54	1350	1.5	4(4)
32	197	−56	1667	1.4	3(4)
33	193	−63	153	4.5	4(4)
34	193	−53	747	2.0	4(4)
35	195	−51	405	2.8	4(4)
36	194	−53	854	1.9	4(4)
37	193	−54	6290	0.7	4(4)
38	195	−55	1124	1.7	4(4)
39	198	−52	755	2.0	4(4)
40	196	−52	1487	1.4	4(4)
41	194	−51	1545	1.4	4(4)
42	201	−51	1250	1.6	3(4)
43	193	−48	1093	1.7	4(4)
44	197	−48	850	1.9	4(4)
45	203	−54	657	2.2	4(4)
46	196	−52	1281	1.5	4(4)
47	202	−52	1334	1.5	4(4)
48	199	−56	1869	1.3	4(4)
49	197	−57	549	2.4	4(4)

TABLE 2. (*concluded*)

Unit number	D(°)	I(°)	k†	α_{95}(°)†	Number used (Number sampled)
50	195	−58	1282	1.5	4(4)
51	194	−57	1273	1.6	4(4)
52	199	−56	361	2.9	4(4)
53	202	−58	1163	1.6	4(4)
54	192	−58	165	4.3	4(4)
55	209	−56	715	2.2	3(4)
56	204	−57	4923	0.8	4(4)
57	208	−60	5000	0.8	3(4)
Section 8					
58	001	+41	2949	1.0	4(4)
Section 9					
59	No satisfactory mean direction				
Section 10					
60	010	+46	285	3.3	4(4)
Section 11					
61	010	+19	1198	1.6	4(4)
62	191	−37	633	2.3	3(4)
Section 12					
63	005	+30	1022	1.7	4(4)
64	006	+30	1038	1.7	4(4)
65	008	+32	1440	1.5	4(4)
66	354	+47	146	4.6	4(4)
67	001	+30	344	3.0	4(4)
Section 13					
68	175	−19	76	6.4	4(4)
69	176	−19	131	4.9	4(4)
70	172	−19	431	2.7	4(4)

†k and α_{95} values are given to indicate the generally very small scatter of the directions that contribute to each unit mean direction. It is considered that this is a useful exercise although clearly such small numbers of vectors cannot have a Fisherian distribution.

(D026: 4.9°, D028: 6.0°, D033: 4.5°, and D054: 4.3°), either declination or inclination values fall well off the smooth curves. This distinction shows that a high within-flow precision is necessary to define the details of secular variation. Where a lower degree of precision is available secular variation can only be approached statistically.

The search for a latitudinal component of plate motion needs a representative average palaeomagnetic direction for the duration of volcanic activity. In the absence of exact knowledge of how individual lavas have sampled the secular variation, there can be no wholly satisfactory way of obtaining this average direction. This general problem is often obscured by the premature application of Fisherian statistics to collected unit mean directions. Single stage averaging in this way is unjustified in the light of a strong suggestion from this study and elsewhere, that eruption frequency has a wider range than has previously been supposed. The nature of this suggestion regarding high variability in eruption frequency is in the common occurrence of sequences of flows having statistically identical palaeomagnetic directions (Wilson 1970*a*; Ade-Hall *et al.* 1974). In some cases the absence of interflow sediments or evidence of interflow weathering supports the implication from the palaeomagnetic data of very high eruption rate compared with the rate of secular variation. For plate motion studies, two-stage averaging of palaeomagnetic data is desirable. The method advocated here is that average directions for sections of flows with identical directions are given unit weight in the calculation of an overall average direction. Each direction contributing to the overall average is

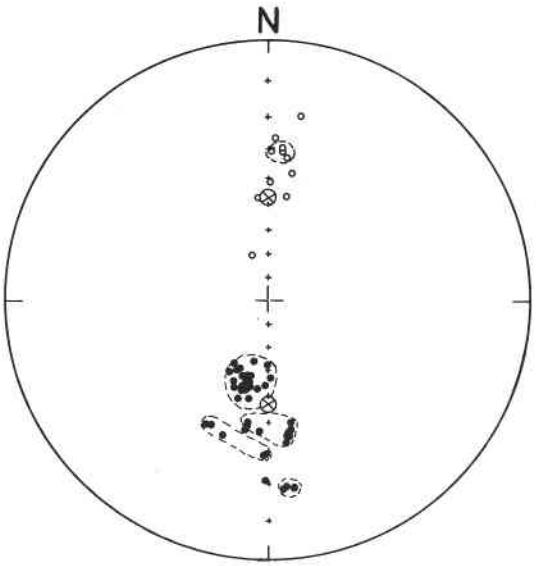


FIG. 2. Mean cleaned remanence directions for individual units. Open circles: lower hemisphere. Filled circles: upper hemisphere. Enclosed groups of directions are from the same sections. Direction for a centered axial dipole field source: ⊗ (equal angle projection).

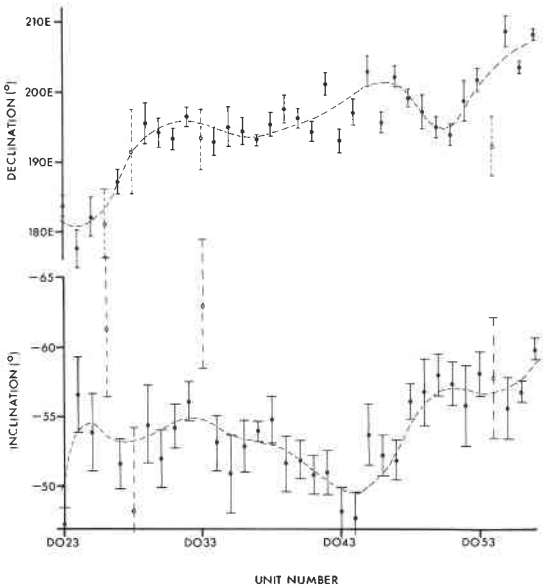


FIG. 3. Section 7, Gleb et Tleah shield volcano, westernmost caldera. Mean cleaned declination and inclination of flows plotted against flow number (D057 youngest). Error bars indicate α_{95} ranges, with broken lines corresponding to α_{95} values in excess of 3°.

termed a temporally independent direction. This method will compensate for repetitious sampling of the field at one point in space and time but has the disadvantage of suppressing evidence for long periods of low secular variation.

Table 3 lists and Fig. 4 illustrates the 14 temporally independent directions determined for the Haruj Assuad volcanics. The final average direction is (N003°E, +38°, α_{95}: 7°) with a corresponding pole position at (83°N, 171°E). This pole is just significantly different from the geographic pole. It is notable that this final average direction is indistinguishable from an average direction obtained excluding contributions from single units. This suggests that both types of temporally independent directions are representatives of the palaeomagnetic field. The difference might be the result of one or more of the following influences: (i) A degree of nonrepresentative sampling of the geomagnetic field, (ii) Plate motion since the initial magnetization of volcanic units, (iii) Offset of the idealized dipole source from the geocenter (Wilson 1970*b*, 1971), and (iv) Polar wander. It is impossible to choose between these influences using data from only one area. However, a *maximum* figure for net plate motion may be estimated. This maximum figure is 7° of latitude (≃800 km) over a period of about 1.0 my, the apparent average age of the Haruj Assuad volcanics, giving an average northwards component of motion of 80 cm y^{-1}. Since this value is two orders of magnitude higher than characteristic plate velocities (or polar wander velocities), the difference between geographic and mean palaeomagnetic poles is most likely to be largely accounted for by mechanisms (i) and (iii). In favor of explanation (i) is the distinctly non-Fisherian distribution of the N, R, and collected temporally independent directions (Fig. 4). It may also be significant that the sense of the difference between the poles agrees with the sense of a worldwide offset described by Wilson (1970*b*) for collected Upper Tertiary data. However, the magnitude of the difference appears to be too large to be accounted for by Wilson's model.

If use is made of the concept of temporally independent palaeomagnetic directions, an esti-

TABLE 3. Mean directions and pole positions for individual sections and for the sections taken together

Section	n	Mean direction D(°)	Mean direction I(°)	k†	α_{95}(°)†	Pole position λ(°)	Pole position φ(°)	δp(°)	δm(°)
1 (flows 1 to 5)	5	194	−33	56	8.4	74N	140E	5.4	9.5
2 (flows 6 & 7)	2	006	+32	76	11.3	78	166	7.2	12.8
3 (flow 8)	1	181	−21	[765]	[2.4]	72	194	[1.3]	[2.6]
4 (flow 9)	1	178	−46	[23]	[11.9]	87	260	[9.7]	[15.2]
5 (flows 10 to 21)	10	177	−37	150	3.6	83	221	2.5	4.3
6 (flow 22)	1	340	+70	[1200]	[1.3]	60	352	[3.0]	[3.5]
7 (flows 23 to 57)	35	195	−54	226	1.6	75	74	1.6	2.2
8 (flow 58)	1	001	+41	[2949]	[1.0]	85	185	[0.8]	[1.2]
9 (plug 59)	1	No consistent cleaned NRM direction							
10 (flow 60)	1	010	+46	[285]	[3.3]	81	110	[2.7]	[4.2]
11 (flow 61)	1	010	+19	[1198]	[1.6]	69	167	[0.9]	[1.7]
(flow 62)	1	191	−37	[633]	[2.3]	77	150	[1.6]	[2.7]
12 (flows 63/4/5/7)	4	005	+31	994	2.2	78	173	[1.4]	[2.5]
13 (flow 66)	1	354	+47	[146]	[4.6]	85	284	[3.8]	[1.9]
14 (flows (?) 68–70)	3	174	−19	2367	1.7	71	216	0.9	1.7
Mean for all temporally independent directions	14	003	+38	27	7.2	83	171	5.0	8.5
Mean of pole positions for temporally independent directions	14	—	—	—	—	84	169		
						(K = 32.6, α_{95} = 6.6)			

†See footnote to Table 2.

mate can be made of the secular variation for the Haruj Assuad volcanic area and the time interval as a whole. Following Doell (1970) and Cox (1969*b*) a palaeoangular standard deviation, S_F of $14.1^{+6.8°}_{-2.9°}$ was obtained for the 14 temporally independent directions listed in Table 3. This value coincides with the expected value for a geomagnetic field model consisting of a geocentric dipole with an 11° wobble on which is superimposed a nondipole field contribution similar to the present one. The relatively large errors associated with this value of S_F reflect the decision to base the estimate on the 14 temporally independent directions rather than the 68 individual unit directions.

Radiometric Dating of the Haruj Assuad Volcanics

Selected volcanic units were dated by the conventional K–Ar method. Apart from some slight alteration of olivine to iddingsite, the samples are fresh, and on the basis of the criteria set forth by Dalrymple and Lamphere (1969), are suitable for K–Ar dating. Prior to splitting for K and Ar analysis, each sample was crushed and sieved to −15/+45 mesh. Potassium concentrations were determined by flame photometry using a Baird–Atomic Model KY-4 instrument. Interference effects were controlled by the lithium internal standard sodium-buffer technique (Cooper 1963). Argon concentrations were determined by isotope dilution using an Associated Electrical Industries type MS10 mass spectrometer. The argon tracer system (a gas pipette) was calibrated by running known amounts of atmospheric argon and also by analyzing interlaboratory standard minerals. Samples of the latter were kindly provided by G. B. Dalrymple and C. T. Harper.

The age of the Haruj Assuad volcanic area appears to range from at least 6.0 to 0.4 my (Table 4). However, the 6.0 my age is for flow or vent material from the Garet Umm el Asol massif that apparently considerably predates the oldest B1 flow age group of Klitzsch (1967). Since most of the temporally independent palaeomagnetic directions are for

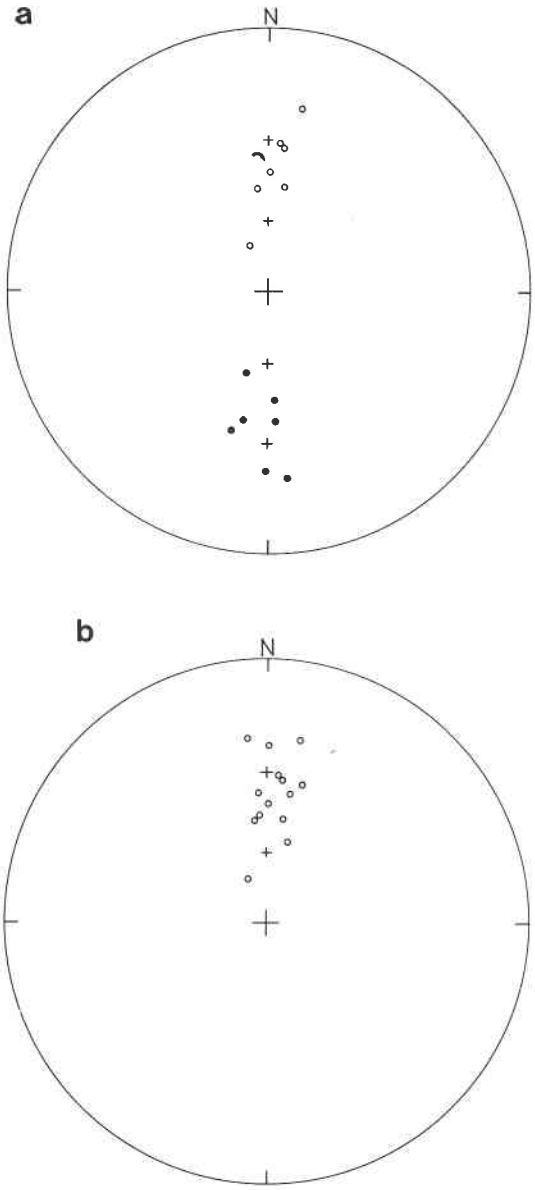


FIG. 4. The fourteen temporally independent directions for the Haruj Assuad volcanics.
4(a) Normal and reverse directions plotted separately.
4(b) All directions as equivalent normal directions (equal angle projection).

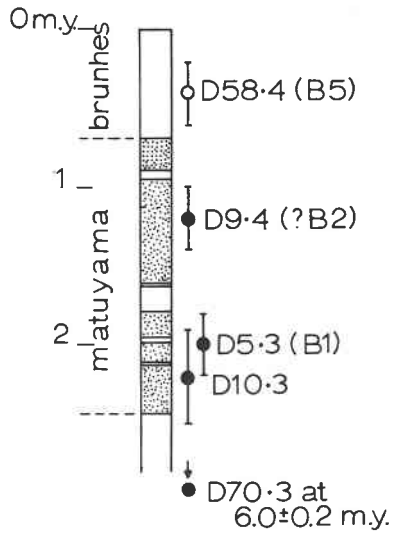


FIG. 5. Whole rock K–Ar ages assigned to their position in the time–polarity scale of Cox (1969). The error bars correspond to ±2 standard durations.

sequences of flows identified with the age group sequence of Klitzsch, the palaeomagnetic data is heavily weighted with representatives in the 2.2 to 0.4 my time interval. This age range coincides closely with the Pliocene–Holocene range proposed by Klitzsch. Older lavas, corresponding in age to the Garet Umm el Asol massif are probably largely unexposed, occurring at depth beneath the younger flows. A satisfactory feature of the radiometric ages is the correspondence of ages and remanence polarities with the time-polarity scale (Cox 1969*a*). Thus, reversely magnetized flow D005 was erupted during the earlier part of the Matuyama reversed epoch. The Garet el Gazia cone most have been one of the sources of the B1 group of flows since both the well defined age of flow D010, from the walls of the cone, is indistinguishable from the age of flow D005 and the reverse mean palaeomagnetic directions for section 1 (flows D001–5) and section 5 (flows D010–21) are very close at (N194°E, −33°, α_{95}: 8.4°) and (N177°E, −37°, α_{95}: 3.6°) respectively. Again, reversely magnetized flow, D009 probably representing age group B2, belongs to the younger part of the Matuyama reversed epoch while normally magnetized flow D058, representing age group B5, falls in the middle of the Brunhes normal epoch. We also note that the age sequence of flows corresponds with the sequence devised by Klitzsch on the basis of surface weathering.

Summary

A combined palaeomagnetic and age-dating study of 70 rock units from the young basaltic

TABLE 4. Whole rock K–Ar ages of Haruj Assuad basalts

Unit	Core	^{40}Ar Radiogenic (scc/g)	$\left[\frac{^{40}Ar \text{ Radiogenic}}{^{40}Ar_{total}}\right]$	$\%K_2O$	Apparent age ±2 standard deviations (my)
D005	3	6.0×10^{-8}	17.5	0.91	2.0 ± 0.2 (Group B1)
D009	4	4.2×10^{-8}	22	1.04	1.2 ± 0.2 (Group B2)
D010	3	4.8×10^{-8}	3	0.66	2.2 ± 0.3
D058	4	1.5×10^{-8}	19	1.10	0.41 ± 0.2 (Group B5)
D061	3	no radiogenic		1.00	<0.9 } (Group B2)
D062	3	argon detected		0.90	<1.8 }
D070	3	3.0×10^{-7}	80	1.50	6.0 ± 0.2

Constants used: $\lambda_e = 0.585 \times 10^{-10}y^{-1}$
$\lambda_\beta = 4.72 \times 10^{-10}y^{-1}$
$^{40}K/K = 1.19 \times 10^{-4}$ mole/mole

Haruj Assuad volcanic area of central Libya has resulted in well defined geophysical parameters for the volcanics. The weighted mean palaeomagnetic pole for volcanics largely in the age range of 2.2 to 0.4 my is at 83°N, 171°E with $\delta p = 5.0°$ and $\delta m = 8.5°$. This pole is just significantly different from the geographic pole. The duration of volcanic activity in the Haruj Assuad area extends back to at least 6.0 my. The reliability of the radiometric ages is confirmed by the complete agreement of the remanence polarities of dated flows with the predictions of the time-polarity scale. A scheme for determining the relative ages of lava groups in the Haruj Assuad area based on the state of weathering of their surfaces is now partly calibrated with absolute ages.

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